$\boxed{L17}$ 13|10|2020

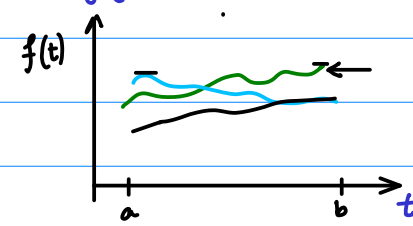
Normed space: Let $\mathcal{V}$ be a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\|\cdot\|$ be a norm on $\mathcal{V}$. Then $\mathcal{V}$ is called a normed space.

e.g. $(\mathbb{R}^n, \|\cdot\|_p)$ is a normed space.

e.g. Bounded continuous function on the interval $[a,b]$ $(\mathcal{C}[a,b])$ is a normed space if we define the following norm

$$\|f\|_\infty := \sup_{t\in[a,b]} |f(t)|$$

$f(t) \in \mathcal{C}[a,b]$



Sequences: $(a_k)_{k=0}^{N-1} = a_0, a_1 \dots a_{N-1}$ $\quad \{a_k\}_{k=0}^{N-1}$

$(a_k)_{k=-\infty}^{\infty}$, $(a_k)_{k=0}^{\infty}$, $\{a_k\}$

Cauchy sequences: Suppose $\mathcal{V}$ (field: $\mathbb{R}$) is a normed space. A sequence $\{v_k\}$ ($v_k \in \mathcal{V}$) is a Cauchy sequence if for each $\varepsilon > 0$, $\varepsilon \in \mathbb{R}$ $\exists M(\varepsilon) \geq 0$ such that $\|v_k - v_\ell\| \leq \varepsilon$ $\forall k, \ell \geq M$.

Intuition: $\|v_k - v_\ell\| \xrightarrow{k,\ell\to\infty} 0$

Appears to converge.

$v_0, v_1, v_2 \dots v_M, v_{M+1}, v_{M+2}, v_{M+3}, \dots \to \infty$

e.g. $\left(\frac{1}{n}\right)_{n=1}^{\infty} \to \{1, 1/2, 1/3, \dots\}$

$\|v_k - v_\ell\| \leq \varepsilon \Rightarrow |1/k - 1/\ell| \leq \varepsilon \quad \forall k, \ell \geq M_\varepsilon$

$k > 0 \Rightarrow 1/k > 0$

$\ell > M_\varepsilon \Rightarrow 1/\ell < 1/M_\varepsilon$

$|1/k - 1/\ell| \leq 1/M_\varepsilon$

If $1/M_\varepsilon < \varepsilon \Rightarrow M_\varepsilon > 1/\varepsilon$

$\varepsilon = 0.1 \qquad M > 1/0.1 \Rightarrow M > 10$

$|1/11 - 1/12| < 0.1 \qquad |1/11 - 1/110| < 0.1$

e.g. $(-1)_{n=1}^{\infty} \to \{-1, 1, -1, 1, \dots\}$ $\quad 0 < \|v_k - v_\ell\| < 2$ ✗ Not a cauchy sequence.

Converging sequence: A sequence $\{a_k\}_{k=0}^{\infty}$ ($a_k \in \mathcal{V}$) converges to L ($\mathcal{V}$) if for each $\varepsilon > 0$ $\exists N \geq 0$ such that $\|a_k - L\| \leq \varepsilon$ $\forall k \geq N$.

$\|a_k - L\| \xrightarrow{k\to\infty} 0$

e.g. $\left(\frac{1}{n}\right)_{n=0}^{\infty}$ $\quad \|1/k - 0\| \leq \varepsilon \quad \forall\, k \geq N_\varepsilon \qquad 1/k \leq 1/N_\varepsilon \leq \varepsilon \Rightarrow N_\varepsilon \geq 1/\varepsilon$

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Real Analysis-I

- Every converging sequence is Cauchy sequence.

However the reverse is not always true.

($\mathbb{F}$ over $\mathbb{F}$)

e.g. $\{1.41, 1.414, 1.4142, 1.41421, \ldots\}$ $\quad (\mathbb{Q}$ over $\mathbb{Q}) \leftarrow$ vector space

$\|x\| = |x| \quad x \in \mathbb{Q}$

- If every Cauchy sequence in the normed space converges then the space is called complete.

e.g. $(\mathbb{R}^n, \mathbb{R}, \|\cdot\|_p)$, $(\mathbb{C}^n, \mathbb{C}, \|\cdot\|_p)$

- A complete normed space is called a Banach space.

Some examples:

① $\ell_p[0,\infty)$ spaces where $1 \leq p \leq \infty$:

For each $1 \leq p \leq \infty$, $\ell_p[0,\infty)$ consists of all sequences $x = (x_0, x_1, \ldots)$ such that $\sum_{i=0}^{\infty} |x_i|^p < \infty$. The associated norm — $\|x\|_p := \left(\sum_{i=0}^{\infty} |x_i|^p\right)^{1/p}$.

② $\ell_\infty[0,\infty)$ space:

$\ell_\infty[0,\infty)$ consists of all bounded sequences $x = (x_0, x_1, \ldots)$ and the ℓ_∞ norm is defined as $\|x\|_\infty := \sup_i |x_i|$.

③ $\mathcal{L}_p(J)$ spaces for $1 \leq p \leq \infty$: $\quad [a,b]$

$\mathcal{L}_1$: $\|f(t)\| = \int_a^b |f(t)| < \infty$ $\quad f(t) \quad \boxed{f(t) \in \mathcal{L}_1[a,b]}$

For each $1 \leq p \leq \infty$, $\mathcal{L}_p(J)$ consists of all Lebesgue measurable functions $x(t)$ defined on the interval $J \subseteq \mathbb{R}$ such that

$$\mathcal{L}_p \rightarrow \|x\|_p := \left(\int_J |x(t)|^p dt\right)^{1/p} < \infty \quad \text{for } 1 \leq p < \infty$$

and $\|x(t)\|_\infty := \operatorname{ess\,sup}_{t \in J} |x(t)|$.